POS System Database

Project Screenshots

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SQL Database Project

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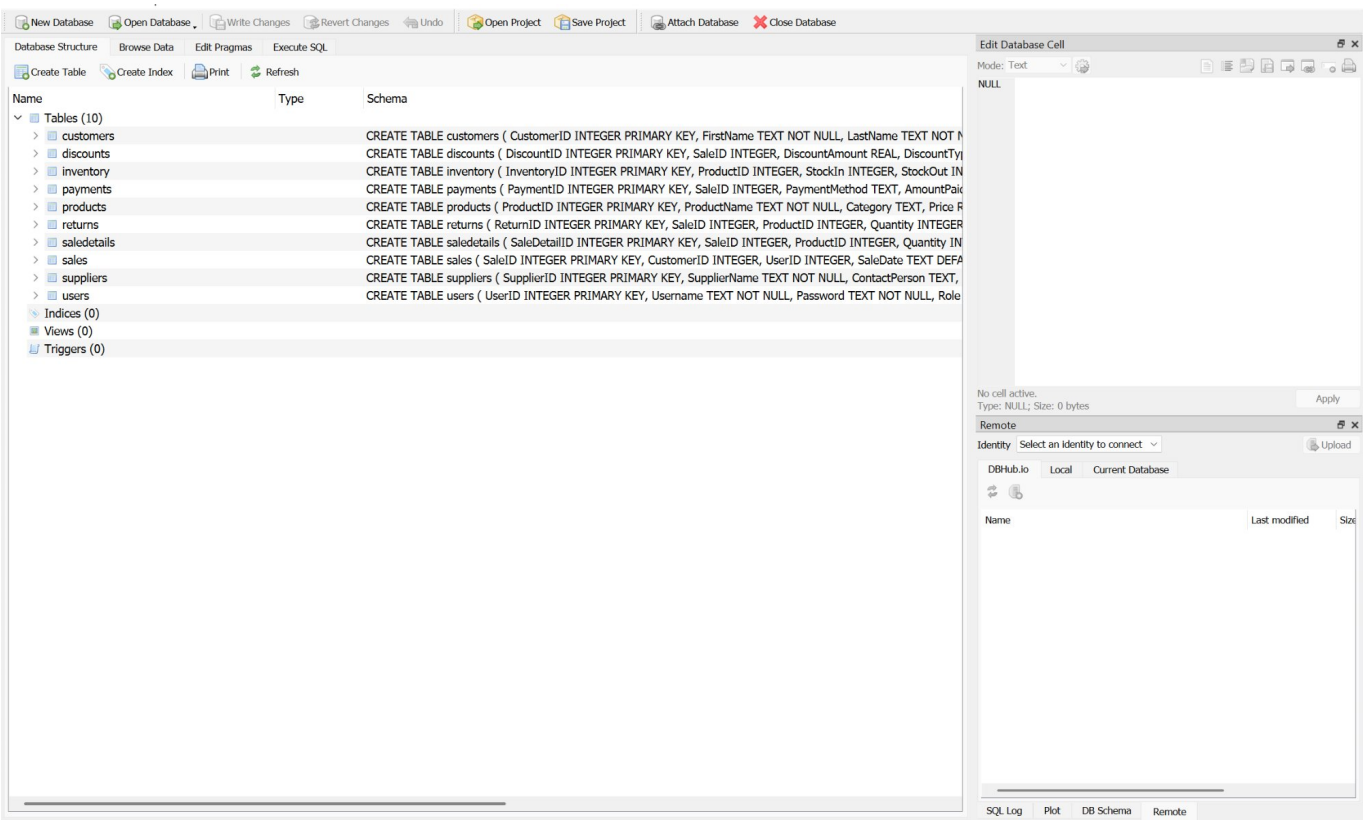
1. Database Structure

All 10 tables in the POS system database



2. Database Setup - SQL Code

The CREATE TABLE and INSERT INTO statements

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Database Setup | Product by Price | Top Customers | Best Sellers | Sales Report

```sql
-- Customers table
CREATE TABLE customers (
  CustomerID INTEGER PRIMARY KEY,
  FirstName TEXT NOT NULL,
  LastName TEXT NOT NULL,
  Email TEXT,
  PhoneNumber TEXT,
  LoyaltyPoints INTEGER DEFAULT 0
);
-- Adding values
INSERT INTO customers VALUES
(1, 'John', 'Doe', 'john.doe@example.com', '1234567890', 100),
(2, 'Jane', 'Smith', 'jane.smith@example.com', '0987654321', 150),
(3, 'Michael', 'Brown', 'michael.b@example.com', '1122334455', 80),
(4, 'Emily', 'Davis', 'emily.d@example.com', '2233445566', 200),
(5, 'Chris', 'Wilson', 'chris.w@example.com', '3344556677', 60),
(6, 'Sarah', 'Johnson', 'sarah.j@example.com', '4455667788', 120),
```

3. Customers Table

Sample customer data with loyalty points

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

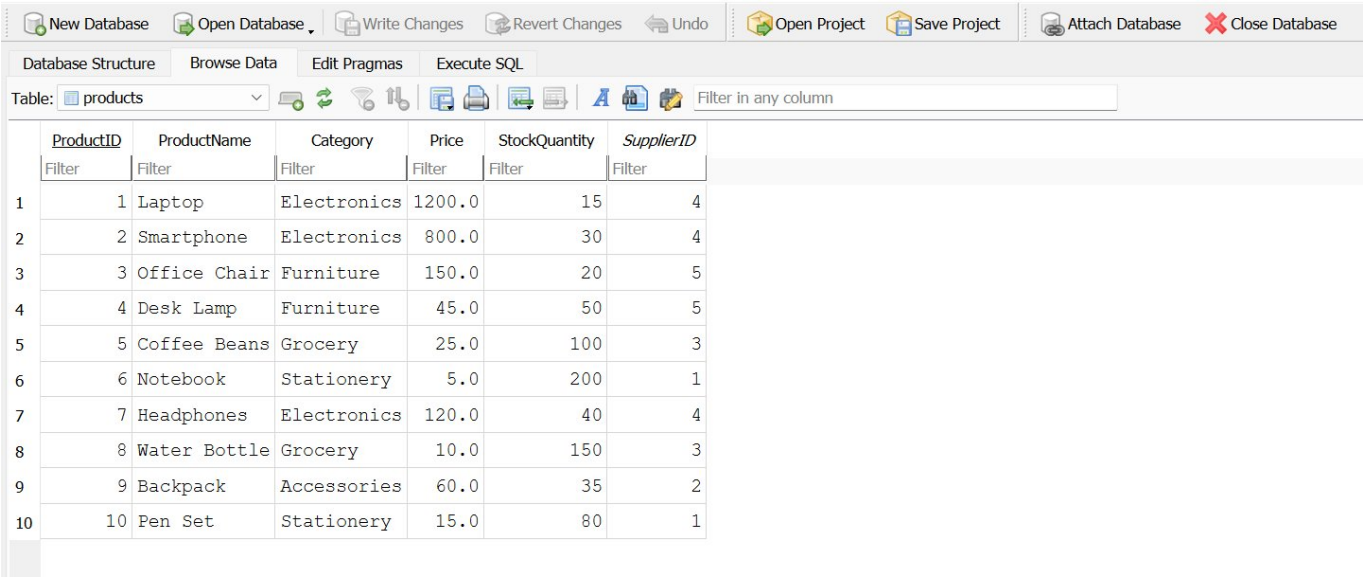
Table: customers

Filter in any column

	CustomerID	FirstName	LastName	Email	PhoneNumber	LoyaltyPoints
	Filter	Filter	Filter	Filter	Filter	Filter
1	1	John	Doe	john.doe@example.com	1234567890	100
2	2	Jane	Smith	jane.smith@example.com	0987654321	150
3	3	Michael	Brown	michael.b@example.com	1122334455	80
4	4	Emily	Davis	emily.d@example.com	2233445566	200
5	5	Chris	Wilson	chris.w@example.com	3344556677	60
6	6	Sarah	Johnson	sarah.j@example.com	4455667788	120
7	7	David	Lee	david.l@example.com	5566778899	90
8	8	Laura	Martin	laura.m@example.com	6677889900	175
9	9	James	Garcia	james.g@example.com	7788990011	50
10	10	Anna	Lopez	anna.l@example.com	8899001122	140

4. Products Table

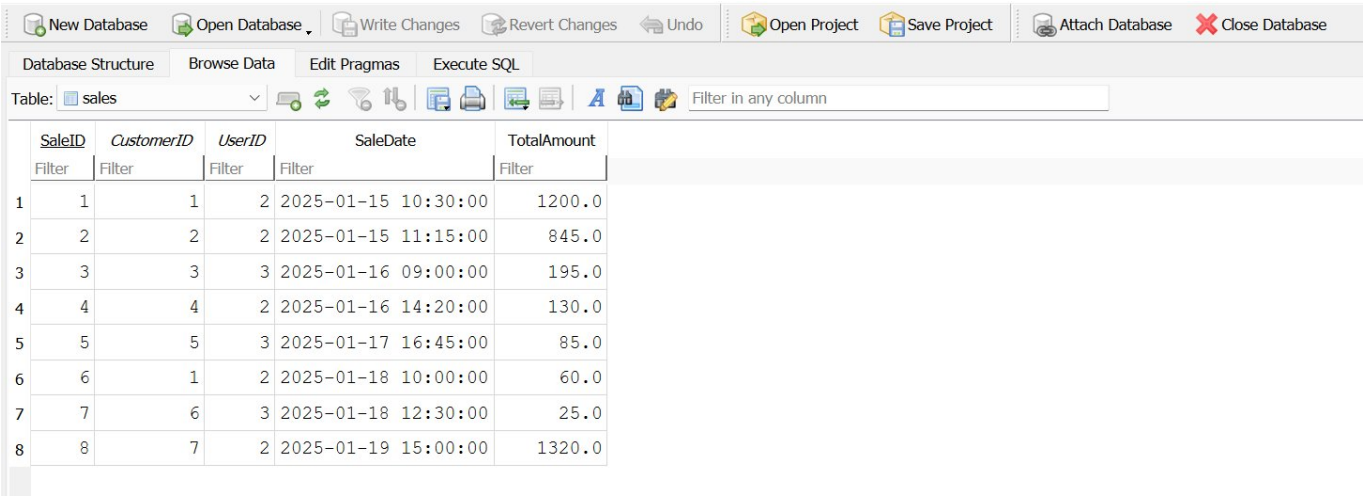
Product catalog with prices and stock quantities



	ProductID	ProductName	Category	Price	StockQuantity	SupplierID
	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Laptop	Electronics	1200.0	15	4
2	2	Smartphone	Electronics	800.0	30	4
3	3	Office Chair	Furniture	150.0	20	5
4	4	Desk Lamp	Furniture	45.0	50	5
5	5	Coffee Beans	Grocery	25.0	100	3
6	6	Notebook	Stationery	5.0	200	1
7	7	Headphones	Electronics	120.0	40	4
8	8	Water Bottle	Grocery	10.0	150	3
9	9	Backpack	Accessories	60.0	35	2
10	10	Pen Set	Stationery	15.0	80	1

5. Sales Table

Sales transaction records



	SaleID	CustomerID	UserID	SaleDate	TotalAmount
	Filter	Filter	Filter	Filter	Filter
1	1	1	2	2025-01-15 10:30:00	1200.0
2	2	2	2	2025-01-15 11:15:00	845.0
3	3	3	3	2025-01-16 09:00:00	195.0
4	4	4	2	2025-01-16 14:20:00	130.0
5	5	5	3	2025-01-17 16:45:00	85.0
6	6	1	2	2025-01-18 10:00:00	60.0
7	7	6	3	2025-01-18 12:30:00	25.0
8	8	7	2	2025-01-19 15:00:00	1320.0

6. Sale Details Table

Line items for each sale (one-to-many relationship)

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

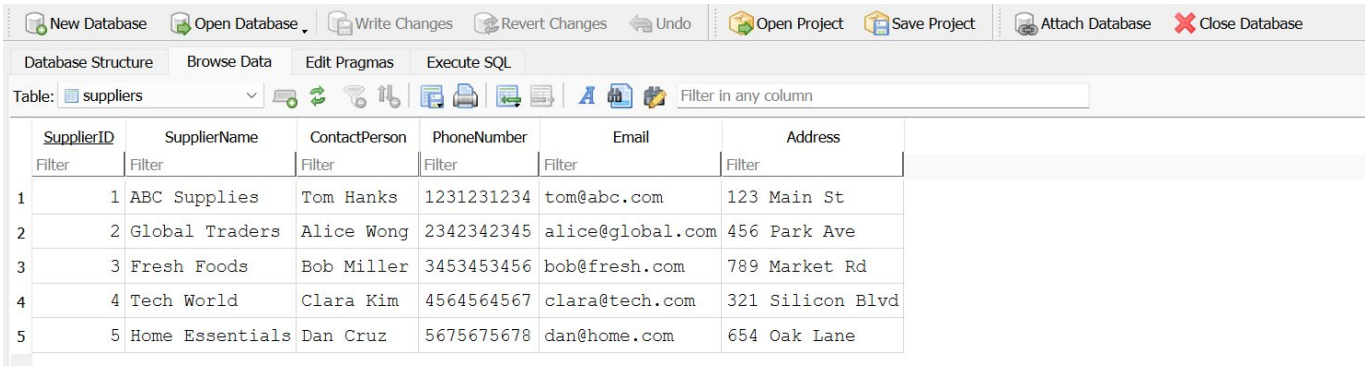
Table: saledetails

Filter in any column

	SaleDetailID	SaleID	ProductID	Quantity	Subtotal
	Filter	Filter	Filter	Filter	Filter
1	1	1	1	1	1200.0
2	2	2	2	1	800.0
3	3	2	6	9	45.0
4	4	3	3	1	150.0
5	5	3	4	1	45.0
6	6	4	7	1	120.0
7	7	4	8	1	10.0
8	8	5	5	2	50.0
9	9	5	10	1	15.0
10	10	5	8	2	20.0
11	11	6	9	1	60.0
12	12	7	5	1	25.0
13	13	8	1	1	1200.0
14	14	8	7	1	120.0

7. Suppliers Table

Supplier contact information



	SupplierID	SupplierName	ContactPerson	PhoneNumber	Email	Address
	Filter	Filter	Filter	Filter	Filter	Filter
1	1	ABC Supplies	Tom Hanks	1231231234	tom@abc.com	123 Main St
2	2	Global Traders	Alice Wong	2342342345	alice@global.com	456 Park Ave
3	3	Fresh Foods	Bob Miller	3453453456	bob@fresh.com	789 Market Rd
4	4	Tech World	Clara Kim	4564564567	clara@tech.com	321 Silicon Blvd
5	5	Home Essentials	Dan Cruz	5675675678	dan@home.com	654 Oak Lane

8. Users Table

Staff accounts (admin, cashiers, manager)

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: users

Filter in any column

	UserID	Username	Password	Role	FullName
	Filter	Filter	Filter	Filter	Filter
1	1	admin	admin123	Admin	System Admin
2	2	cashier1	pass123	Cashier	Mary Jones
3	3	cashier2	pass456	Cashier	Peter Parker
4	4	manager1	mgr789	Manager	Bruce Wayne

9. Query 1 - Products Sorted by Price

SELECT with ORDER BY - products from most to least expensive

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Database Setup | Product by Price | Top Customers | Best Sellers | Sales Report

```sql
/*Product sorted by price*/

SELECT * FROM products ORDER BY Price DESC;
```

	ProductID	ProductName	Category	Price	StockQuantity	SupplierID
1	1	Laptop	Electronics	1200.0	15	4
2	2	Smartphone	Electronics	800.0	30	4
3	3	Office Chair	Furniture	150.0	20	5
4	7	Headphones	Electronics	120.0	40	4
5	9	Backpack	Accessories	60.0	35	2
6	4	Desk Lamp	Furniture	45.0	50	5
7	5	Coffee Beans	Grocery	25.0	100	3
8	10	Pen Set	Stationery	15.0	80	1
9	8	Water Bottle	Grocery	10.0	150	3
10	6	Notebook	Stationery	5.0	200	1

```
Execution finished without errors.
Result: 10 rows returned in 39ms
At line 1:
/*Product sorted by price*/
SELECT * FROM products ORDER BY Price DESC;
```

10. Query 2 - Top Customers by Spending

JOIN + GROUP BY + aggregation (COUNT, SUM)

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Database Setup | Product by Price | Top Customers | Best Sellers | Sales Report

```sql
/* Top customers by spending */

SELECT c.FirstName, c.LastName,
       COUNT(s.SaleID) AS TotalOrders,
       SUM(s.TotalAmount) AS TotalSpent
FROM customers c
JOIN sales s ON c.CustomerID = s.CustomerID
GROUP BY c.CustomerID
ORDER BY TotalSpent DESC;
```

	FirstName	LastName	TotalOrders	TotalSpent
1	David	Lee	1	1320.0
2	John	Doe	2	1260.0
3	Jane	Smith	1	845.0
4	Michael	Brown	1	195.0
5	Emily	Davis	1	130.0
6	Chris	Wilson	1	85.0
7	Sarah	Johnson	1	25.0

```
Execution finished without errors.
Result: 7 rows returned in 41ms
At line 1:
/* Top customers by spending */

SELECT c.FirstName, c.LastName,
       COUNT(s.SaleID) AS TotalOrders,
       SUM(s.TotalAmount) AS TotalSpent
FROM customers c
JOIN sales s ON c.CustomerID = s.CustomerID
GROUP BY c.CustomerID
ORDER BY TotalSpent DESC;
```

11. Query 3 - Best Selling Products

JOIN between products and saledetails with SUM

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Database Setup | Product by Price | Top Customers | Best Sellers | Sales Report

```sql
/* Best selling products */

SELECT p.ProductName, SUM(sd.Quantity) AS TotalSold
FROM products p
JOIN saledetails sd ON p.ProductID = sd.ProductID
GROUP BY p.ProductID
ORDER BY TotalSold DESC;
```

	ProductName	TotalSold
1	Notebook	9
2	Water Bottle	3
3	Coffee Beans	3
4	Headphones	2
5	Laptop	2
6	Pen Set	1
7	Backpack	1
8	Desk Lamp	1
9	Office Chair	1
10	Smartphone	1

```
Execution finished without errors.
Result: 10 rows returned in 27ms
At line 1:
/* Best selling products */

SELECT p.ProductName, SUM(sd.Quantity) AS TotalSold
FROM products p
JOIN saledetails sd ON p.ProductID = sd.ProductID
GROUP BY p.ProductID
ORDER BY TotalSold DESC;
```

12. Query 4 - Full Sales Report

Three-table JOIN with string concatenation

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Database Setup | Product by Price | Top Customers | Best Sellers | Sales Report*

```sql
/*  Sales report with customer and cashier */

SELECT s.SaleID,
       c.FirstName || ' ' || c.LastName AS Customer,
       u.FullName AS Cashier,
       s.TotalAmount,
       s.SaleDate
FROM sales s
JOIN customers c ON s.CustomerID = c.CustomerID
JOIN users u ON s.UserID = u.UserID;
```

	SaleID	Customer	Cashier	TotalAmount	SaleDate
1	1	John Doe	Mary Jones	1200.0	2025-01-15 10:30:00
2	2	Jane Smith	Mary Jones	845.0	2025-01-15 11:15:00
3	3	Michael Brown	Peter Parker	195.0	2025-01-16 09:00:00
4	4	Emily Davis	Mary Jones	130.0	2025-01-16 14:20:00
5	5	Chris Wilson	Peter Parker	85.0	2025-01-17 16:45:00
6	6	John Doe	Mary Jones	60.0	2025-01-18 10:00:00
7	7	Sarah Johnson	Peter Parker	25.0	2025-01-18 12:30:00
8	8	David Lee	Mary Jones	1320.0	2025-01-19 15:00:00

```
Execution finished without errors.
Result: 8 rows returned in 34ms
At line 1:
/* Best selling products */

SELECT s.SaleID,
       c.FirstName || ' ' || c.LastName AS Customer,
       u.FullName AS Cashier,
       s.TotalAmount,
       s.SaleDate
FROM sales s
JOIN customers c ON s.CustomerID = c.CustomerID
JOIN users u ON s.UserID = u.UserID;
```